

Machine Learning

Maths Problem Sets 1–4: Problems Only

Computations, derivations and short proofs for Lectures 2–4. No essay questions.

How to use these sets

These four sets are *computation and derivation only*: every question asks you to calculate a number, derive a formula, or prove a short result. They follow the lectures: Set 1 is decisions under a known distribution (Lecture 2, first half), Set 2 is learning from data (Lecture 2, second half), Set 3 is generalisation (Lecture 3), Set 4 is discriminative and generative classification (Lecture 4).

Each problem opens with a blue box holding the shared setup, then a yellow box listing the formulas you need with a pointer [one-read §x] to where they are explained in the one-read document. Each sub-question sits in its own bar, so you can work it out on paper first. The same sets with full solutions are in *ML_Maths_Problem_Sets_Solutions.pdf*. Difficulty: ★ direct application, ★★ several steps, ★★★ derivation or proof. Problems marked *looking ahead* use a derivative that later lectures on optimisation will need; they are good practice now.

All logarithms are natural ($\ln$) unless written $\log_2$. Where a normal-distribution value is needed, Φ denotes the standard normal CDF and the values needed are given.

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1 Set 1: Decisions under a known distribution

1.1 Problem 1.1: Three actions, three classes

Problem 1.1 (★)

An intrusion detector gives each flow posterior probabilities for three classes: *normal*, *minor attack*, *major attack*. Three actions are available: *ignore*, *monitor*, *block*. The cost $c(a, y)$ of action a when the truth is y is:

	normal	minor	major
ignore	0	5	20
monitor	1	0	8
block	4	2	0

Tools you need

Conditional risk $\mathbb{E}[\ell_{\text{dec}}(Y, a) \mid X = x] = \sum_y c(a, y) P(Y = y \mid x)$ and the Bayes rule: take the action with the smallest conditional risk [one-read §2.2].

(a) For posteriors $(0.5, 0.3, 0.2)$, compute the conditional risk of each action and the optimal action.

(b) Now let the posteriors be $(0.8 - q, 0.2, q)$ with $0 \leq q \leq 0.8$. Write the three conditional risks as functions of q and find the optimal action for every q .

(c) Suppose the actions are the classes themselves and $c(a, y) = \mathbf{1}(a \neq y)$. Show that the Bayes rule chooses the most probable class.

1.2 Problem 1.2: The pinball loss and a uniform outcome

Problem 1.2 (★★)

For a fixed input x , the output is uniform: $Y \mid x \sim \text{Uniform}[2, 10]$, density $\frac{1}{8}$ on $[2, 10]$. The asymmetric (“pinball”) loss charges $c_u = 3$ per unit of under-prediction and $c_o = 1$ per unit of

over-prediction:

$$\ell(y, a) = \begin{cases} 3(y - a), & y > a, \\ 1(a - y), & y \leq a. \end{cases}$$

Tools you need

Squared loss $\rightarrow$ conditional mean, absolute loss $\rightarrow$ conditional median, pinball loss $\rightarrow$ quantile at level $c_u/(c_u + c_o)$ [one-read §2.2].

(a) Give the optimal prediction under squared loss and under absolute loss.

(b) For $2 \leq a \leq 10$, show that $\mathbb{E}[(Y - a)_+] = (10 - a)^2/16$ and $\mathbb{E}[(a - Y)_+] = (a - 2)^2/16$, where $u_+ = \max(u, 0)$.

(c) Minimise the expected pinball loss directly and check that you get the quantile predicted by the general formula.

(d) Compute the expected pinball loss at $a = 8$ and at the mean $a = 6$. How much does reporting the mean cost?

1.3 Problem 1.3: Every summary is optimal for its own loss

Problem 1.3 (★★)

For a fixed input x , the output takes two values: $P(Y = 10 \mid x) = 0.3$ and $P(Y = 0 \mid x) = 0.7$.

Tools you need

Expectation and variance of a discrete variable; the squared-loss decomposition [one-read §2.2, mastery box].

(a) Show that for any constant a , $\mathbb{E}[(Y - a)^2 \mid x] = \text{Var}(Y \mid x) + (\mathbb{E}[Y \mid x] - a)^2$.

(b) Compute $\mathbb{E}[Y \mid x]$, $\text{Var}(Y \mid x)$, and the expected squared loss at $a = 3$, $a = 5$ and $a = 0$.

(c) Find the median, and the expected absolute loss at $a = 0$ and at $a = 3$.

(d) Summarise what (b) and (c) show.

1.4 Problem 1.4: The reject option in general, and what review is worth

Problem 1.4 (★★★)

Binary detection with $p = P(Y = 1 \mid x)$, false-positive cost C_{FP} , false-negative cost C_{FN} , and a review action of fixed cost C_R after which the case is resolved correctly. Correct automatic decisions cost 0.

Tools you need

Conditional risks $R(\text{benign}) = C_{FN} p$, $R(\text{attack}) = C_{FP}(1 - p)$, $R(\text{review}) = C_R$; the lower envelope of the risk lines [one-read §2.2].

(a) Without review, show that the risk of the best action, $\min\{C_{FN}p, C_{FP}(1-p)\}$, is largest at $p^* = C_{FP}/(C_{FP} + C_{FN})$, with value $C_{FP}C_{FN}/(C_{FP} + C_{FN})$.

(b) Deduce that review is optimal for some p if and only if $C_R < C_{FP}C_{FN}/(C_{FP} + C_{FN})$, and that the regions are then: benign for $p \leq C_R/C_{FN}$, review in between, attack for $p \geq 1 - C_R/C_{FP}$.

(c) Apply this to $C_{FP} = 3$, $C_{FN} = 12$, $C_R = 1.5$.

(d) Suppose the posteriors of incoming flows are uniformly distributed on $[0, 1]$. Compute the expected cost per flow of the optimal policy with and without review.

1.5 Problem 1.5: Bayes error of two Gaussian classes

Problem 1.5 (★★)

One feature: $X \mid (Y = 0) \sim \mathcal{N}(0, 1)$ and $X \mid (Y = 1) \sim \mathcal{N}(2, 1)$. Values you may need: $\Phi(1) = 0.8413$, $\Phi(1.693) = 0.9548$, $\Phi(0.307) = 0.6205$, and $\Phi(-z) = 1 - \Phi(z)$.

Tools you need

Bayes classifier: compare $\pi_c p(x \mid c)$ [one-read §2.2, 4.3]; the error of a threshold rule is a sum of two normal tail probabilities.

(a) With equal priors, find the Bayes decision rule.

(b) Compute its error (the Bayes error).

(c) Now $\pi_0 = 0.8$, $\pi_1 = 0.2$. Find the new boundary.

(d) Compute the Bayes error with these priors, and the error if you kept the boundary $x = 1$.

2 Set 2: Learning from data

2.1 Problem 2.1: KNN by hand, and leave-one-out

Problem 2.1 (★)

Six training points, each with a numerical output y (for regression) and a class c (for classification):

x	1	2	4	5	7	8
y	2.0	3.0	5.0	4.0	8.0	9.0
c	0	0	1	1	0	1

The query is $x = 4.6$; distance is $|x - x'|$.

Tools you need

$\hat{f}_k(x) = \frac{1}{k} \sum_{i \in N_k(x)} y^{(i)}$ and $\hat{P}_k(c \mid x) = \frac{1}{k} \sum_{i \in N_k(x)} \mathbf{1}(c^{(i)} = c)$ [one-read §2.4].

(a) Order the training points by distance to the query, and give the KNN regression prediction for $k = 1, 3, 5, 6$.

(b) Give $\hat{P}_k(c = 1 \mid 4.6)$ and the zero-one decision for the same k .

(c) Compute the leave-one-out (LOO) error of 1-NN classification and the LOO mean squared error of 1-NN regression: predict each point from the other five. Why is the ordinary training error of 1-NN zero?

2.2 Problem 2.2: Which distance?

Problem 2.2 (★)

Query $x = (0, 0)$; training points $A = (3, 0)$ and $B = (2, 2)$. The Minkowski distance is $d_p(x, z) = (\sum_j |x_j - z_j|^p)^{1/p}$.

Tools you need

Minkowski family: $p = 1$ Manhattan, $p = 2$ Euclidean [one-read §2.4].

(a) Which point is nearer under $p = 1$, and under $p = 2$?

(b) Show that $d_p(x, B) \rightarrow 2 = \max_j |x_j - B_j|$ as $p \rightarrow \infty$, and say which point is nearer in that limit.

2.3 Problem 2.3: Softmax, cross-entropy and their derivatives

Problem 2.3 (★★), parts (c) and (d) looking ahead

A three-class model outputs scores $s = (s_1, s_2, s_3) = (2, 1, -1)$.

Tools you need

$p_c = e^{s_c} / \sum_k e^{s_k}$, $\ell_{CE} = -\ln p_y$, $\sigma(s) = 1/(1 + e^{-s})$ [one-read §2.5, 4.2].

(a) Compute the softmax probabilities, and the cross-entropy loss if the true class is $y = 2$.

(b) Show that subtracting 2 from every score leaves the probabilities unchanged, and that for two classes softmax is the sigmoid of the score difference.

(c) Show that $\partial \ell_{CE} / \partial s_c = p_c - \mathbf{1}(c = y)$, and evaluate this gradient for (a).

(d) Binary case with $y \in \{0, 1\}$ and $\hat{p} = \sigma(s)$: show that $\ell_{BCE} = \ln(1 + e^{-s})$ when $y = 1$ and that $\partial \ell_{BCE} / \partial s = \sigma(s) - y$. Evaluate the loss and the derivative at $s = 3$ and $s = -3$ for $y = 1$.

2.4 Problem 2.4: Entropy, KL divergence and mutual information

Problem 2.4 (★★)

Work in bits. $p = (0.5, 0.25, 0.25)$ and $q = (0.7, 0.2, 0.1)$. Values you may need: $\log_2 0.7 = -0.5146$, $\log_2 0.2 = -2.3219$, $\log_2 0.1 = -3.3219$, $\log_2 0.3 = -1.7370$, $\log_2 0.8 = -0.3219$.

Tools you need

$H(p) = -\sum p_c \log p_c$, $H(p, q) = -\sum p_c \log q_c$, $D_{KL}(p||q) = \sum p_c \log \frac{p_c}{q_c}$, $H(p, q) = H(p) + D_{KL}(p||q)$, $I(X; Y) = H(Y) - H(Y | X)$ [one-read §2.5].

(a) Compute $H(p)$, $H(p, q)$ and $D_{KL}(p||q)$, and check the identity $H(p, q) = H(p) + D_{KL}(p||q)$.

(b) Compute $D_{KL}(q||p)$. Is the divergence symmetric?

(c) $X, Y \in \{0, 1\}$ with joint probabilities $P(0, 0) = P(1, 1) = 0.4$ and $P(0, 1) = P(1, 0) = 0.1$. Compute $H(Y)$, $H(Y | X)$, $I(X; Y)$, the Bayes error, and the smallest achievable expected cross-entropy.

(d) A model predicts $\hat{p}(Y = 1 | X = 1) = 0.7$ and $\hat{p}(Y = 1 | X = 0) = 0.3$. Compute its expected cross-entropy with the decomposition $R_{CE} = H(Y | X) + \mathbb{E}_X[D_{KL}]$, then directly.

2.5 Problem 2.5: ERM in closed form

Problem 2.5 (★★)

Empirical risk minimisation for very simple model families, where the minimiser can be written down.

Tools you need

$\hat{\theta} \in \arg \min_{\theta} \hat{R}_{\text{train}}(\theta)$; set the derivative of the training risk to zero [one-read §2.6].

(a) A constant predictor $f(x) = c$ is fitted to $y = (1, 2, 2, 3, 12)$. Show that squared loss gives the sample mean and absolute loss the sample median. Compute both, their training risks, and each one's risk under the other loss.

(b) A constant probability $q = \hat{p}(y = 1)$ is fitted with cross-entropy to N binary labels containing n_1 ones. Show that $\hat{q} = n_1/N$, and compute the minimal training risk for $n_1 = 3$, $N = 10$.

(c) Fit $f(x) = wx$ (no intercept) by squared loss. Show $\hat{w} = \sum_i x_i y_i / \sum_i x_i^2$ and compute it, with the training MSE, for the data $(1, 2), (2, 3), (3, 7)$.

(d) Now add an L_2 penalty: minimise $\frac{1}{N} \sum (y_i - wx_i)^2 + \lambda w^2$. Show $\hat{w}_{\lambda} = \sum x_i y_i / (\sum x_i^2 + N\lambda)$ and compute it for $\lambda = 1$ with its training MSE.

3 Set 3: Generalisation

3.1 Problem 3.1: Bias and variance of a shrinkage estimator

Problem 3.1 (★★)

Observations $Y_i = \theta + \varepsilon_i$, $i = 1, \dots, N$, with independent noise of mean 0 and variance σ^2 . To predict a new Y , use $\hat{\theta}_\alpha = \alpha \bar{Y}$ with $0 \leq \alpha \leq 1$, where $\bar{Y} = \frac{1}{N} \sum_i Y_i$. The case $\alpha = 1$ is the sample mean; $\alpha < 1$ shrinks it toward 0, like a penalty.

Tools you need

Bias, variance and $\text{MSE} = \text{bias}^2 + \text{variance}$ [one-read §3.3]; $\mathbb{E}\bar{Y} = \theta$, $\text{Var}\bar{Y} = \sigma^2/N$.

- (a) Compute the bias and the variance of $\hat{\theta}_\alpha$.
- (b) Show that $\text{MSE}(\alpha) = \mathbb{E}[(\hat{\theta}_\alpha - \theta)^2] = (1 - \alpha)^2 \theta^2 + \alpha^2 \sigma^2 / N$, and find the α^* that minimises it.
- (c) With $\theta = 2$, $\sigma^2 = 4$, $N = 4$: compute α^* , $\text{MSE}(1)$, $\text{MSE}(\alpha^*)$ and $\text{MSE}(0.5)$.
- (d) Show that the expected squared error in predicting a new, independent $Y = \theta + \varepsilon$ is $\sigma^2 + \text{MSE}(\alpha)$, and evaluate it at $\alpha = 1$ and α^* .
- (e) Why can α^* not be computed in practice, and how would it be chosen?

3.2 Problem 3.2: Bias and variance of KNN regression

Problem 3.2 (★★)

True function $f^*(x) = x^2$; observations $y_i = f^*(x_i) + \varepsilon_i$ with independent noise of variance $\sigma^2 = 0.01$, at fixed inputs $x = \pm 0.1, \pm 0.2, \pm 0.3, \pm 0.4, \pm 0.5$ (one observation each). Query: $x = 0$, where $f^*(0) = 0$. Only the noise is random (the inputs are fixed).

Tools you need

$\hat{f}_k(0) = \frac{1}{k} \sum_{i \in N_k(0)} y_i$; bias = $\mathbb{E}\hat{f}_k(0) - f^*(0)$; variance of an average of k independent noises is σ^2/k [one-read §3.3].

- (a) For even k , show that $\text{Bias}(0) = \frac{1}{k} \sum_{i \in N_k(0)} x_i^2$ and $\text{Var}(0) = \sigma^2/k$.
- (b) Compute bias, variance and $\text{MSE} = \text{bias}^2 + \text{variance}$ for $k = 2, 4, 6, 8, 10$, and find the best k .
- (c) What is the expected squared prediction error for a new observation at $x = 0$ with the best k ?

3.3 Problem 3.3: How precise is a test score?

Problem 3.3 (★★)

A *fixed* classifier has true error rate R (zero-one loss). It is evaluated on N independent test examples. Chebyshev's inequality: for any variable Z with mean μ and variance v , $P(|Z - \mu| \geq t) \leq v/t^2$.

Tools you need

For a fixed model, $\mathbb{E}\hat{R} = R$ and $\text{Var } \hat{R} = \sigma_\ell^2/N$ [one-read §2.6, mastery box].

- (a) Show that $\hat{R}_{\text{test}}$ is unbiased with variance $R(1 - R)/N$.
- (b) For $R = 0.08$ and $N = 2000$, compute the standard error and an approximate 95% interval (two standard errors).
- (c) How many test examples make the standard error at most 0.002?
- (d) A recall of 0.9 is measured on a test set containing 50 positives. What is its standard error?
- (e) Use Chebyshev to bound $P(|\hat{R}_{\text{test}} - R| \geq 0.02)$ in the setting of (b).

3.4 Problem 3.4: Selection noise

Problem 3.4 (★★)

Validation errors are $\hat{R}_j = R_j + \varepsilon_j$ with independent $\varepsilon_j \sim \mathcal{N}(0, \sigma^2)$. You may use: $\mathbb{E} \min(Z_1, Z_2) = -1/\sqrt{\pi}$ and $\mathbb{E} \min(Z_1, Z_2, Z_3) = -3/(2\sqrt{\pi})$ for independent standard normals; $\Phi(0.354) = 0.638$, $\Phi(1.645) = 0.95$.

Tools you need

Winner's curse $\mathbb{E} \min_j \hat{R}_j \leq \min_j R_j$ [one-read §3.1, 3.5]; the difference of two independent normals.

- (a) Three configurations all have $R = 0.20$ and $\sigma = 0.02$. What is the expected validation error reported for the selected one?
- (b) Two configurations have true errors 0.20 and 0.21, with $\sigma = 0.02$ each. What is the probability of selecting the worse one?
- (c) Take $\sigma^2 \approx 0.16/N$ for both, where N is the validation-set size ($\sigma = 0.02$ is $N = 400$). How large must N be for the probability in (b) to fall below 5%?

3.5 Problem 3.5: L_1 and L_2 in numbers

Problem 3.5 (★★)

One-dimensional penalised problems with a least-squares estimate z :

$$\text{lasso: } \min_{\theta} \frac{1}{2}(\theta - z)^2 + \lambda|\theta|, \quad \text{ridge: } \min_{\theta} \frac{1}{2}(\theta - z)^2 + \lambda\theta^2.$$

Tools you need

Regularised ERM, L_1 versus L_2 [one-read §3.4]; for the lasso, treat $\theta > 0$, $\theta < 0$ and $\theta = 0$ separately.

(a) Show that the lasso solution is $\text{sign}(z) \max(|z| - \lambda, 0)$ and the ridge solution is $z/(1 + 2\lambda)$. Compute both for $z = 0.8$ with $\lambda = 0.5$ and $\lambda = 1$.

(b) With an orthonormal design each coefficient decouples. For least-squares coefficients $z = (3, -0.5, 1.2, 0.1)$ and $\lambda = 0.6$, compute both solutions and count the non-zero coefficients.

(c) Constrained form: minimise $(\theta_1 - 2)^2 + (\theta_2 - 1)^2$ subject to $|\theta_1| + |\theta_2| \leq 1$, then subject to $\theta_1^2 + \theta_2^2 \leq 1$.

3.6 Problem 3.6: Early stopping and patience**Problem 3.6 (★)**

Validation loss after each epoch of training (at most 12 epochs):

epoch	1	2	3	4	5	6	7	8	9	10	11	12
loss	.90	.70	.58	.52	.50	.51	.53	.52	.49	.47	.46	.48

Rule: stop after p consecutive epochs without improving on the best loss so far; return the best checkpoint.

Tools you need

$t^* = \arg \min_t \hat{R}_{\text{val}}(\hat{\theta}_t)$ with patience [one-read §3.4].

(a) For patience $p = 2, 3, 4$: when does training stop, which checkpoint is returned, and how many epochs were run?

(b) Interpret the result.

3.7 Problem 3.7: Leakage through a scaler**Problem 3.7 (★)**

A one-feature anomaly detector standardises its input and flags $|z| > 3$. Training values: 2, 4, 6, 8. A test value: 20. Standard deviations use the $1/N$ formula.

Tools you need

$\hat{\mu}_j, \hat{\sigma}_j$ fitted on training data only [one-read §3.6].

(a) Fit the scaler on the training data, transform the test value, and decide.

- (b) Now fit the scaler on all five values (the leaky way). Transform the test value and decide.

3.8 Problem 3.8: Precision depends on the base rate

Problem 3.8 (★★)

A detector has recall (true-positive rate) 0.9 and false-positive rate 0.02, whatever the prevalence π of attacks.

Tools you need

Recall $TP/(TP+FN)$, precision $TP/(TP+FP)$, false-positive rate $FP/(FP+TN)$, F_1 [one-read §3.6]; Bayes' rule.

(a) Show that precision =
$$\frac{\text{Rec} \cdot \pi}{\text{Rec} \cdot \pi + \text{FPR} \cdot (1 - \pi)}.$$

- (b) Compute precision, F_1 and the number of alarms per 100,000 flows for $\pi = 0.1, 0.01, 0.001$.

- (c) What false-positive rate is needed for a precision of 0.5 at $\pi = 0.001$?

4 Set 4: Discriminative and generative classification

4.1 Problem 4.1: Logistic regression by hand

Problem 4.1 (★★), part (e) looking ahead

A logistic regression model with $w = (1.5, -2)$ and $b = 0.5$: $s(x) = w^\top x + b$, $\hat{p}(y = 1 \mid x) = \sigma(s(x))$.

Tools you need

Sigmoid, cross-entropy, log-odds, probability thresholds as score thresholds [one-read §4.2]; $\partial \ell_{BCE} / \partial s = \sigma(s) - y$ (Problem 2.3).

- (a) Compute the score and the probability for $x_A = (1, 1)$ and $x_B = (2, 0.5)$.

- (b) The labels are $y_A = 1$ and $y_B = 0$. Compute both cross-entropy losses.

- (c) Show that the log-odds $\ln \frac{\hat{p}}{1-\hat{p}}$ equal the score, and deduce what a one-unit increase in x_1 does to the odds.

- (d) Write the decision boundary for a cost threshold of 0.8 on $\hat{p}$.

- (e) Take one gradient step on x_B alone with learning rate 0.1: use $\nabla_w \ell = (\sigma(s) - y)x$ and $\partial \ell / \partial b = \sigma(s) - y$. What is the new probability for x_B ?

4.2 Problem 4.2: Maximum likelihood for Gaussian classes

Problem 4.2 (★★★)

One class with observations $x_1, \dots, x_N$ modelled as $\mathcal{N}(\mu, \sigma^2)$, and C classes with counts $N_1, \dots, N_C$ ($\sum_c N_c = N$) for the priors.

Tools you need

Log-likelihood, setting derivatives to zero, a Lagrange multiplier for $\sum_c \pi_c = 1$ [one-read §4.3, 4.4].

(a) Derive $\hat{\mu} = \frac{1}{N} \sum_i x_i$ and $\hat{\sigma}^2 = \frac{1}{N} \sum_i (x_i - \hat{\mu})^2$.

(b) Show that maximising $\sum_c N_c \ln \pi_c$ subject to $\sum_c \pi_c = 1$ gives $\hat{\pi}_c = N_c/N$.

(c) Show that $\mathbb{E}[\hat{\sigma}^2] = \frac{N-1}{N} \sigma^2$ when the x_i are independent $\mathcal{N}(\mu, \sigma^2)$. Hint: first show $\sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - N(\bar{x} - \mu)^2$.

(d) For a class with observations 2, 4, 9, compute $\hat{\mu}$, the ML variance, and the unbiased variance.

4.3 Problem 4.3: LDA and QDA disagree

Problem 4.3 (★★★)

Two classes in $\mathbb{R}^2$, four points each:

class 0: (0, 0), (2, 0), (0, 2), (2, 2); class 1: (4, 4), (6, 4), (5, 5), (5, 3).

Use ML estimates. The pooled covariance for LDA is $\hat{\Sigma} = \sum_c \frac{N_c}{N} \hat{\Sigma}_c$. Query point: $x = (3, 3)$.

Tools you need

Gaussian ML estimates, QDA score $-\frac{1}{2}(x - \hat{\mu}_c)^\top \hat{\Sigma}_c^{-1}(x - \hat{\mu}_c) - \frac{1}{2} \ln |\hat{\Sigma}_c| + \ln \hat{\pi}_c$, LDA score $x^\top \hat{\Sigma}^{-1} \hat{\mu}_c - \frac{1}{2} \hat{\mu}_c^\top \hat{\Sigma}^{-1} \hat{\mu}_c + \ln \hat{\pi}_c$ [one-read §4.4].

(a) Compute $\hat{\pi}_c$, $\hat{\mu}_c$, $\hat{\Sigma}_c$ and the pooled $\hat{\Sigma}$.

(b) QDA: compute both scores at (3, 3), the prediction and $\hat{p}(y = 1 | x)$.

(c) LDA: compute both scores at (3, 3) (the common $\ln 0.5$ may be dropped), the prediction and $\hat{p}(y = 1 | x)$.

(d) Find both decision boundaries and explain the disagreement.

(e) Count the parameters estimated by LDA and by QDA for $d = 2$, $C = 2$, and for $d = 50$, $C = 3$.

4.4 Problem 4.4: Gaussian Naive Bayes, and why variances need smoothing

Problem 4.4 (★★)

Two features (log-duration x_1 , log-bytes x_2), independent given the class:

	$x_1 \mid y$	$x_2 \mid y$	prior
benign	$\mathcal{N}(3, 1)$	$\mathcal{N}(6, 4)$	0.9
attack	$\mathcal{N}(1, 0.25)$	$\mathcal{N}(4, 1)$	0.1

Use $\ln \mathcal{N}(x; \mu, \sigma^2) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}$ and $\ln(2\pi) = 1.8379$.

Tools you need

Naive Bayes score $\ln \hat{\pi}_c + \sum_j \ln p(x_j \mid c)$; variance smoothing $\tilde{\sigma}^2 = \hat{\sigma}^2 + \lambda s^2$ [one-read §4.5].

(a) For $x = (1.5, 5)$, compute every log-density, both class scores, and $\hat{p}(\text{attack} \mid x)$.

(b) Suppose the fitted attack variance of x_1 were 0.001 (a nearly constant feature in a tiny sample). Compute $\ln p(x_1 = 1.2 \mid \text{attack})$. Then smooth it with $\lambda = 0.1$ and an overall variance $s^2 = 2$, and recompute.

4.5 Problem 4.5: Multinomial Naive Bayes for spam

Problem 4.5 (★★)

Vocabulary (login, password, verify, invoice, meeting). Training token counts: spam (8, 6, 5, 1, 0), total 20; ham (1, 1, 0, 4, 9), total 15. Priors: spam 0.4, ham 0.6. New message: “verify password password meeting”, count vector $x = (0, 2, 1, 0, 1)$.

Tools you need

$$p(x \mid c) \propto \prod_j \theta_{cj}^{x_j}, \hat{\theta}_{cj} = \frac{n_{cj} + \alpha}{\sum_\ell n_{c\ell} + \alpha V} \text{ [one-read §4.5].}$$

(a) With $\alpha = 1$, compute the smoothed token probabilities of each class.

(b) Compute both log-scores and $\hat{p}(\text{spam} \mid x)$.

(c) What happens without smoothing?

(d) What does the posterior tend to as $\alpha \rightarrow \infty$?

4.6 Problem 4.6: Prior shift in odds form

Problem 4.6 (★★)

A binary classifier was trained on data resampled to balance the classes ($\pi_t = 0.5$). In deployment, attacks are $\pi_d = 0.02$ of flows. Assume prior shift and a calibrated training posterior $r_t(x)$.

Tools you need

Posterior odds = prior odds $\times$ likelihood ratio; the prior-shift correction [one-read §4.7].

(a) Derive $\frac{r_d}{1 - r_d} = \frac{\pi_d/(1 - \pi_d)}{\pi_t/(1 - \pi_t)} \cdot \frac{r_t}{1 - r_t}$.

(b) Compute r_d for $r_t = 0.5$ and $r_t = 0.9$.

(c) In deployment, alerts are raised when $r_d > 0.2$. What threshold on r_t is equivalent?

(d) A generative model gives likelihood ratio $\Lambda(x) = 6$ for some flow. Compute its posterior under $\pi = 0.5$ and $\pi = 0.02$, and check against (a).

4.7 Problem 4.7: Confident but unfamiliar

Problem 4.7 (★★)

Two classes on one feature, equal priors: $X | (Y = 0) \sim \mathcal{N}(0, 1)$, $X | (Y = 1) \sim \mathcal{N}(4, 1)$. A reject rule accepts the predicted class c^* only if $\ln p(x | c^*) \geq \tau$. Use $\ln(2\pi) = 1.8379$; 99% of a standard normal lies within ± 2.576 .

Tools you need

Posterior from class scores; the reject option $\ln p(x | c^*) \geq \tau_{c^*}$ [one-read §4.8].

(a) Compute $\hat{p}(y = 1 | x = 9)$.

(b) Choose τ so that 99% of each class's own points are accepted.

(c) Is $x = 9$ accepted? And $x = 2$?